Python

Figures taken or adapted from Horstmann, Necaise 2019
Python for Everyone, 3/e, Wiley.



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String Operations

Figures taken or adapted from Horstmann, Necaise 2019 Python for Everyone, 3/e, Wiley.

Statement	Result	Comment
`string = "Py"` `string = string + "thon"`	string is set to "Python"	When applied to strings, + denotes concatenation.
`print("Please" +` `" enter your name: ")`	Please enter your name:	Use concatenation to break up strings that don't fit into one line.
`team = str(49) + "ers"`	team is set to "49ers"	Because 49 is an integer, it must be converted to a string.
`greeting = "H & S"` `n = len(greeting)`	n is set to 5	Each space counts as one character.
`string = "Sally"` `ch = string[1]`	ch is set to "a"	Note that the initial position is 0.
`last = string[len(string) - 1]`	last is set to the string containing the last character in string	The last character has position len(string) - 1.

Adapted from Horstmann, Necaise *Python for Everyone*, 3/e

Numerical Division

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Expression (where n = 1729)	Value		Comment
`n % 10`	9	Last digit	For any positive integer *n*, `n % 10` is the last digit of *n*.
`n // 10`	172	*n* without last digit	This is *n* without the last digit.
`n % 100`	29	Last two digits	The last two digits of *n*.
`n % 2`	1	1 if odd 0 if even	`n % 2` is 0 if *n* is even, 1 if *n* is odd (provided *n* is not negative).
`-n // 10`	-173	Truncates toward zero (no floor)	−173 is the largest integer ≤ −172.9. We will not use floor division for negative numbers in this book.

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Math Functions

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Function	Returns
`abs`(x)	The absolute value of x.
`round`(x) `round`$(x,\ n)$	The floating-point value x rounded to a whole number or to n decimal places.
`max`$(x_1,\ x_2,\ \ldots,\ x_n)$	The largest value from among the arguments.
`min`$(x_1,\ x_2,\ \ldots,\ x_n)$	The smallest value from among the arguments.

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String Formatting Identifiers

Figures taken or adapted from Horstmann, Necaise 2019 Python for Everyone, 3/e, Wiley.

Syntax print(*formatString* % (*value*$_1$, *value*$_2$, ..., *value*$_n$))

```
print( "Quantity: %d  Total: %10.2f" % (quantity, total) )
```

Format string
Text to display, including format specifiers.

Format operator
Applies the values to the format string.

Values
Variables, constants, or expressions to insert into the format string.

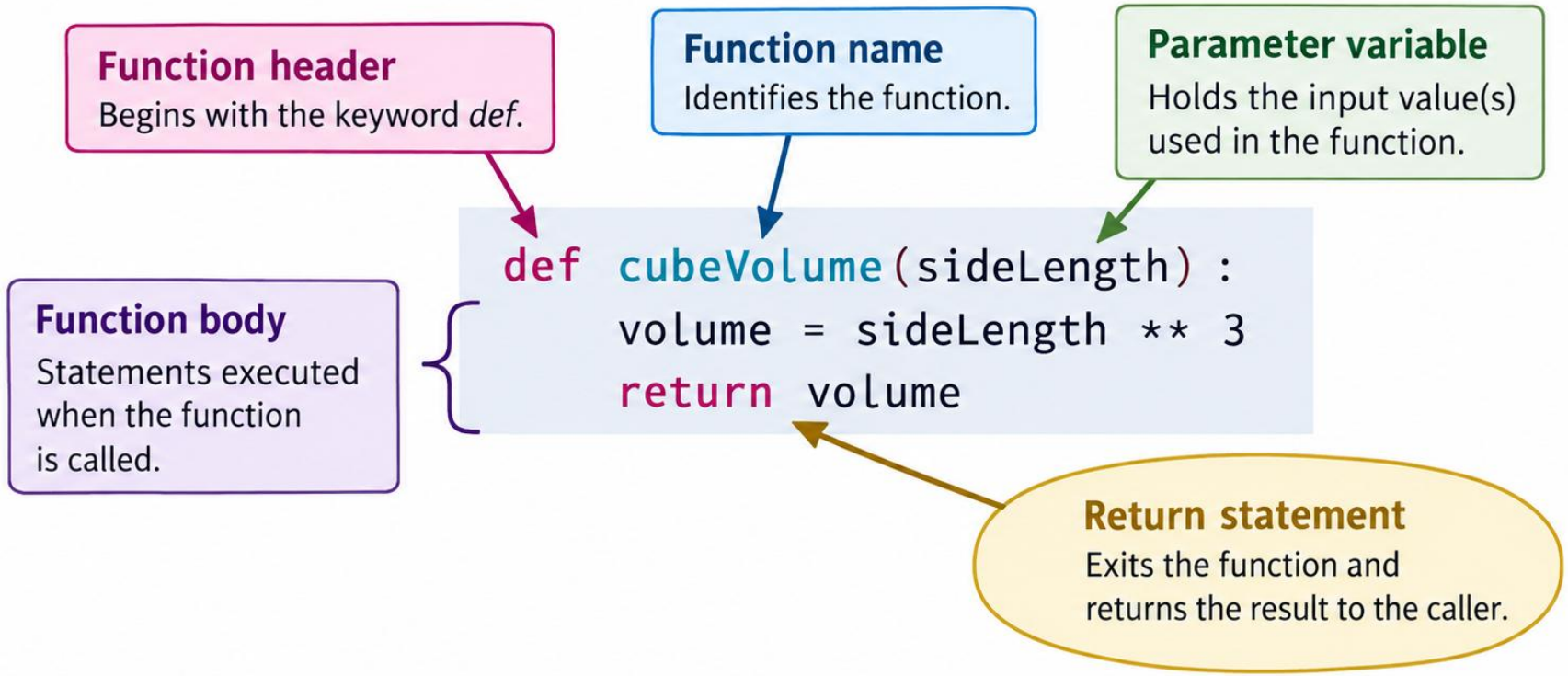
Format String	Sample Output	Comments
"%d"	24	Use d with an integer.
"%5d"	␣␣␣24	Spaces are added so that the field width is 5.
"%05d"	00024	If you add 0 before the field width, zeros are added instead of spaces.
"Quantity:%5d"	Quantity:␣␣␣24	Characters inside a format string but outside a format specifier appear in the output.
"%f"	1.219970	Use f with a floating-point number.
"%.2f"	1.22	Prints two digits after the decimal point.
"%7.2f"	␣␣␣1.22	Spaces are added so that the field width is 7.
"%s"	Hello	Use s with a string.
"%d %.2f"	24␣1.22	You can format multiple values at once.
"%9s"	␣␣␣␣Hello	Strings are right-justified by default.
"%-9s"	Hello␣␣␣␣	Use a negative field width to left-justify.
"%d%%"	24%	To add a percent sign to the output, use %%.

Functions Syntax

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Syntax

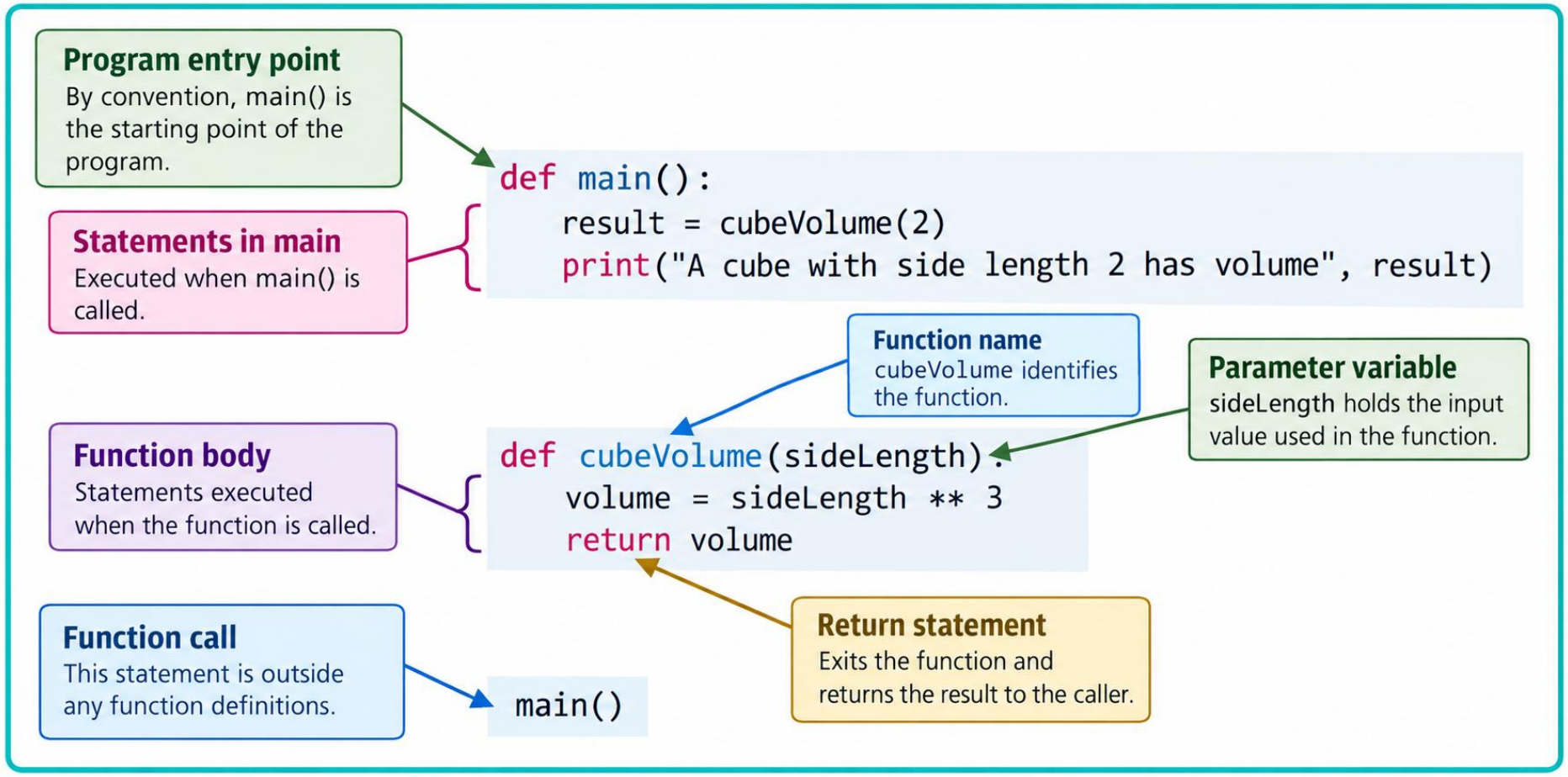
```
def functionName(parameterName1, parameterName2, . . . , parameterN):
    statements
```



Adapted from Horstmann, Necaise *Python for Everyone*, 3/e

Functions in a Program Main

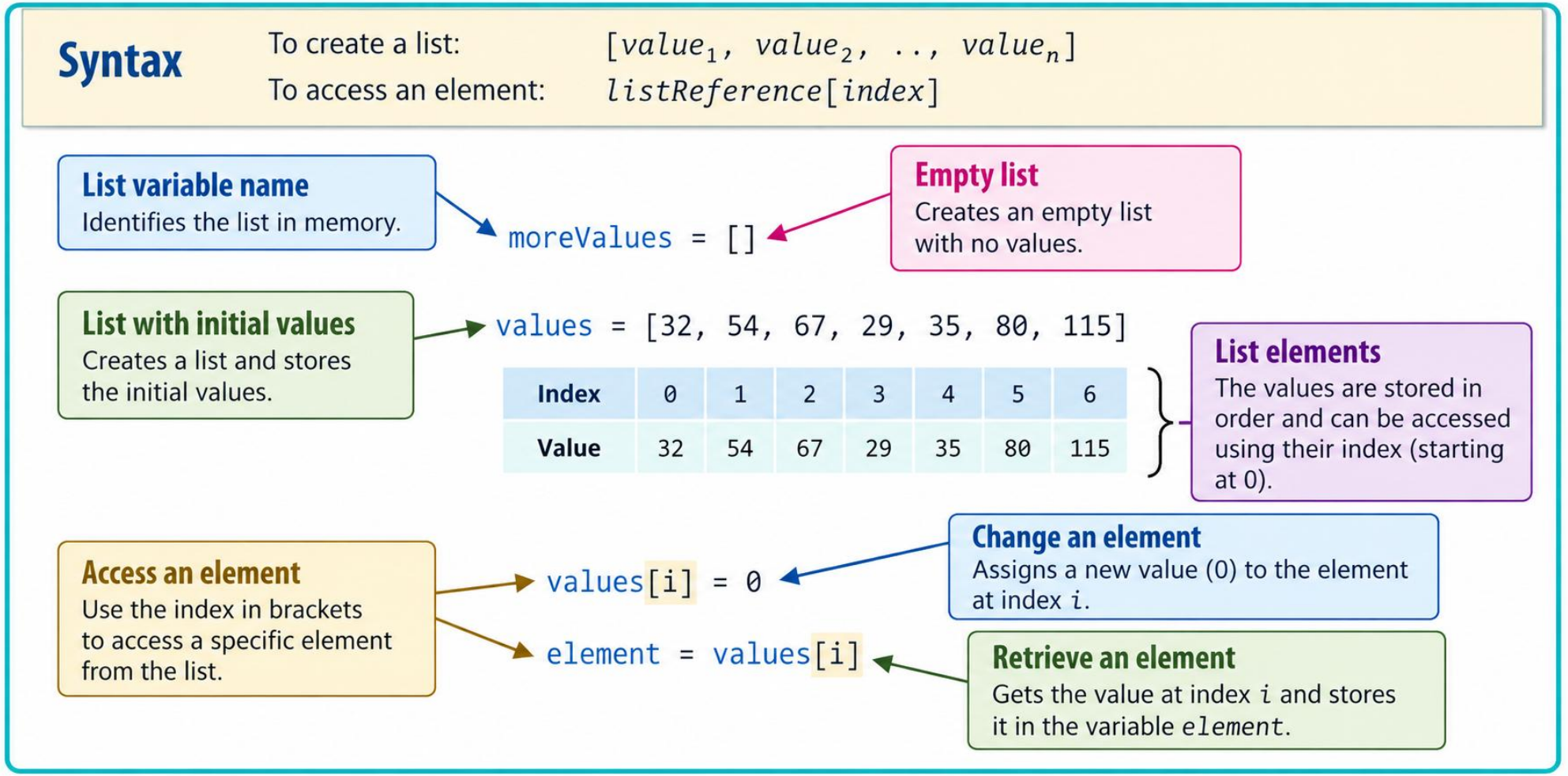
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Lists Syntax

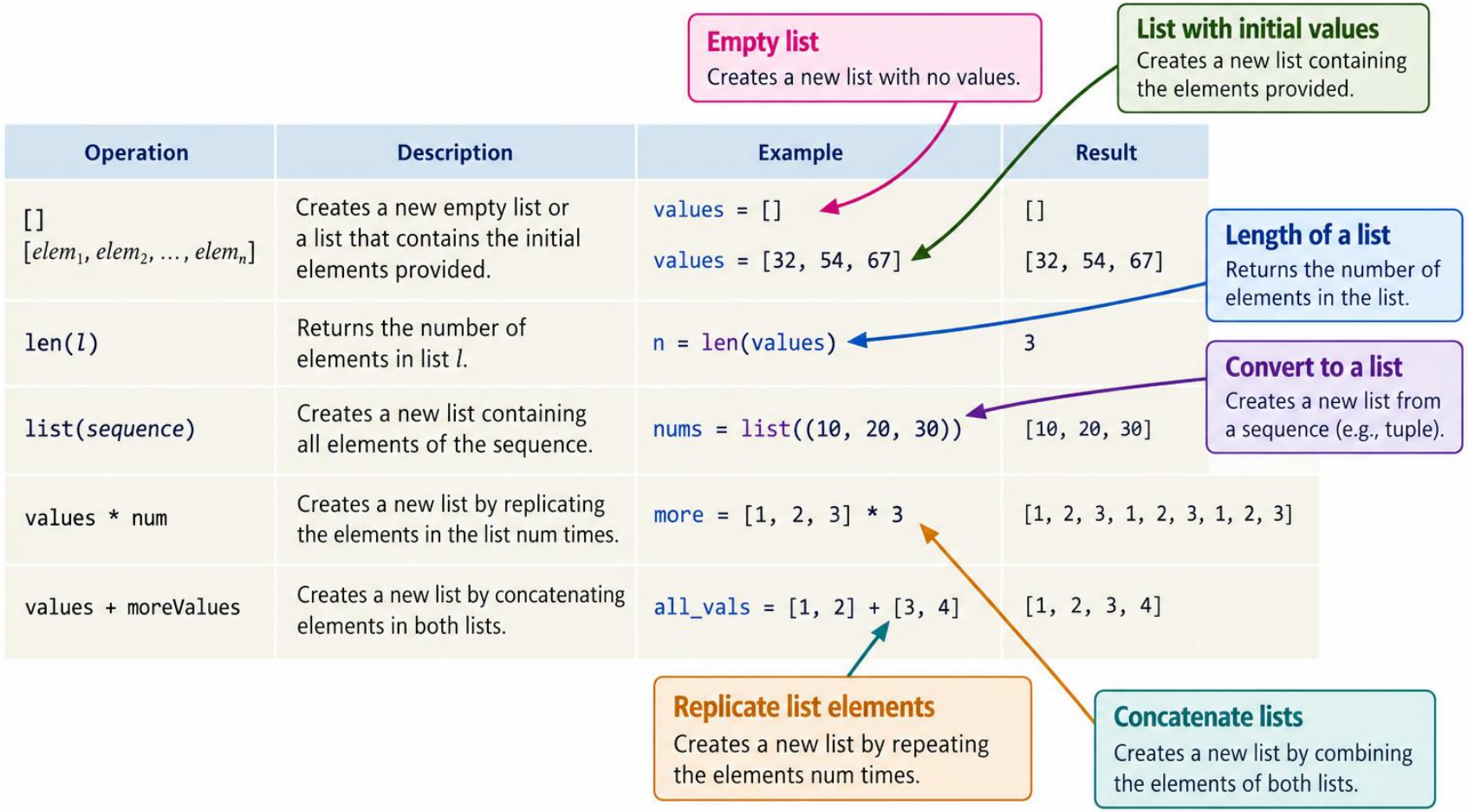
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Lists Functions / operators

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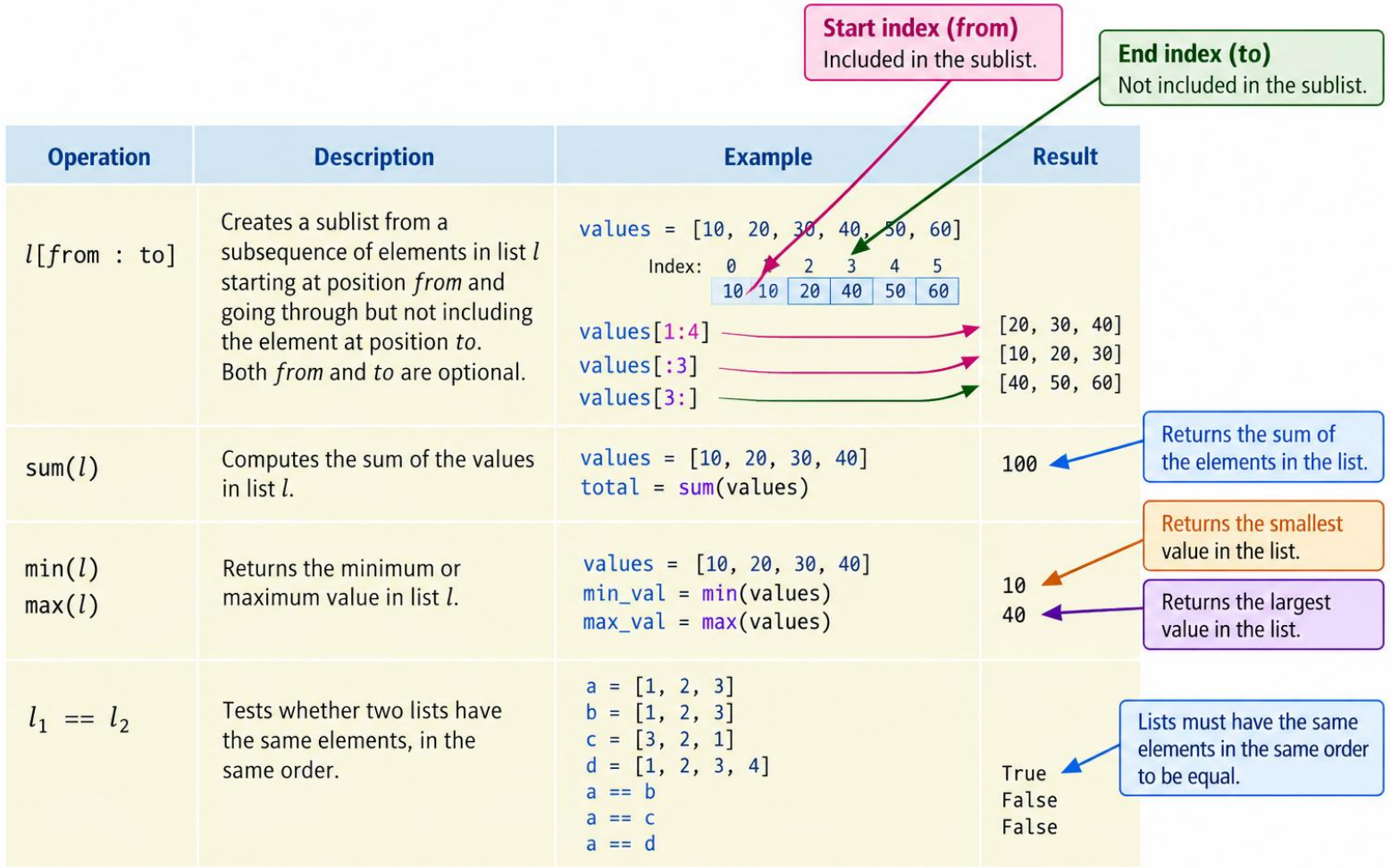


Operation	Description	Example	Result
`[]` `[`*elem*$_1$, *elem*$_2$, …, *elem*$_n$`]`	Creates a new empty list or a list that contains the initial elements provided.	`values = []` `values = [32, 54, 67]`	`[]` `[32, 54, 67]`
`len(l)`	Returns the number of elements in list *l*.	`n = len(values)`	`3`
`list(sequence)`	Creates a new list containing all elements of the sequence.	`nums = list((10, 20, 30))`	`[10, 20, 30]`
`values * num`	Creates a new list by replicating the elements in the list num times.	`more = [1, 2, 3] * 3`	`[1, 2, 3, 1, 2, 3, 1, 2, 3]`
`values + moreValues`	Creates a new list by concatenating elements in both lists.	`all_vals = [1, 2] + [3, 4]`	`[1, 2, 3, 4]`

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Lists Functions / operators

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Lists Methods

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Method	Description	Example (starting with `l = [10, 20, 30, 40]`)	Result

`l.pop()`
`l.pop(position)`
	Removes the last element from the list or from the given position. All elements following the given position are moved up one place.	`l = [10, 20, 30, 40]`
		`x = l.pop()` ← Remove last element.
		`y = l.pop(1)` ← Remove element at position 1.
			`x = 40`
			`l = [10, 20, 30]`
			`y = 20`
			`l = [10, 30]`

`l.insert(position, element)`	Inserts the element at the given position in the list. All elements at and following the given position are moved down.	`l = [10, 20, 30, 40]` `l.insert(2, 99)` ← Insert 99 at position 2.	`l = [10, 20, 99, 30, 40]`
`l.append(element)`	Appends the element to the end of the list.	`l = [10, 20, 30, 40]` `l.append(50)` ← Add 50 to end of list.	`l = [10, 20, 30, 40, 50]`
`l.index(element)`	Returns the position of the given element in the list. The element must be in the list.	`l = [10, 20, 30, 40]` `pos = l.index(30)` ← Returns the position of 30.	`pos = 2`
`l.remove(element)`	Removes the given element from the list and moves all elements following it up one position.	`l = [10, 20, 30, 40]` `l.remove(30)` ← Remove the value 30 from the list.	`l = [10, 20, 40]`
`l.sort()`	Sorts the elements in the list from smallest to largest.	`l = [40, 10, 30, 20]` `l.sort()` ← Sorts the list in ascending order.	`l = [10, 20, 30, 40]`

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